

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q21-1692

Strawberry Hill 138 kV  
September 30, 2022

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1692. The Interconnection Customer proposes a 57.0 MW solar photovoltaic facility interconnecting at the Strawberry Hill 138 kV substation in the Pennsylvania area. The request was received on March 11, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 484 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,431,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                       | Contingency               | Loading (%) | Rating (MVA) | DFAX (%) |
|----------------------------------------------------------|---------------------------|-------------|--------------|----------|
| Fourth Branch Hydroelectric Facility – White Heath 138kV | Elk River – Mercury 345kV | 108.5       | 159          | 18.3     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                                 | Required Upgrade           | Cost Estimate | Allocation to Project |
|----------------------------------------------------------|----------------------------|---------------|-----------------------|
| Fourth Branch Hydroelectric Facility – White Heath 138kV | Reconductor line           | \$4,259,000   | 34% (\$1,431,000)     |
|                                                          | Total allocated to project |               | \$1,431,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.